

6. Technical Description / Development Platform

6.1 Overview of Technical Approach

SavePlate will be developed as a web-based application that will be using full-stack development and architecture. Our frontend will provide a responsive user interface accessible across device and while our backend will handle all of the important things like business logic, data persistence and notification management. This project will be a follow of iterative and incremental development methodologies as defined in the project schedule.

6.2 Development Platform

Documentation & Modelling

Criteria	Tool / Software	Version	Description
Documentation	Microsoft Word	2019/365	It will be used to write all of the format of all project documentation including proposal, requirements spec and used to design documents.
Modelling – Req.	Draw.io / Lucidchart	Free tier	It will be used to create Use Case Diagram as well as system flow models.
Modelling – Design	Draw.io	Free tier	It will be used to create ERD, Data Dictionary diagram and Architecture Diagrams.
UI Prototyping	Figma	Free tier	It will be used to design our website wireframes and user interface mockups for all of the use case.
Project Management	GanttProject	3.3.3322	It will be used to create and manage the project like Gantt Chart and resource allocations.
Collaboration	GitHub	Latest	We will use GitHub for version control and team collaboration repository and to host all source code and documentation.
Communication	WhatsApp / Discord	Latest	It will be used for communication, meeting coordination and progress updates.

Framework & Development

Criteria	Tool / Software	Version	Description
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Frontend	HTML5, CSS3, JavaScript / React.js	React 18+	We will use this modern framework for frontend and it will be responsible for responsive web UI built with component based React framework for the process of data rendering.
Backend	Node.js with Express.js	Node 20 LTS	For Backend the RESTful API server will be handling all of the business, logics, authentications and database operations.
Database	MySQL / MariaDB	8.0+	Relational Database will be used for storing user account, food inventory, donations, meal plans and for analytics data.
Authentication	JWT + bcrypt + TOTP (2FA)	Latest	JSON web tokens will be responsible for authentication like session management, b-crypt for password hashing and we will use TOTP for 2-factor authentications.
Notification	NodeMailer / Web Push API	Latest	Our web application will also have function like email notifications for expiry alerts; and browser push notifications for the real time updates.
Testing	Jest + Selenium	Latest	For Testing we will use unit and integrations like testing with Jest; functional / UI testing with using Selenium Web Driver.

Hardware Specifications (Development Notebook)

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 (or equivalent) minimum quad-core
RAM	8 GB minimum (16 GB recommended for running local dev server + database)
Storage	256 GB SSD minimum (for project files, node_modules, and database storage)
Operating System	Windows 10/11, macOS 12+, or Ubuntu 20.04 LTS and Other Linux
Display	13-inch minimum for development; Full HD (1920×1080) recommended.
Internet Connection	Stable broadband will be required for GitHub pushes and online collaboration.

Demonstration Platform

Criteria	Tool	Version	Description
Browser	Google Chrome	Latest stable	Primary demo browser, supports all HTML5, CSS3, and JavaScript will be used in the app.

Browser (Alt)	Mozilla Firefox	Latest stable	Be used by Fallback Demo browser for cross browser compatibility verifications.
Local Server	localhost (Node.js)	Port 3000	Application will be run locally for this development machine during the demonstrations.
Database	MySQL Workbench	8.0+	It will be used to demonstrate data persistence and for query execution during UAT.

7. Risk Management Plan

The following risk management tables will show the SavePlate project that identifies the possible risks and checks how likely they are to happen and how serious their effect could be and creates plan to prevent them and handle them if they occur as a team.

Risk Name	Description	Prob.	Mitigation Strategy
Team Member Unavailability	Team Member may be not present due to illness, personal problem or any other academic pressure that causes delay to assigned tasks.	M	There will be redistribution of task immediately among the remaining members and there will be shared documentation on GitHub so any member can continue another member work.
Scope Creep	There might be Additional feature that may be requested or assumed during the development, expanding the project beyond its defined boundaries that might cause schedule overrun.	M	We will stick to 6 agreed features and if any changes team need it will be discussed with team leader's approval. Then, we will finalize and lock the project scope.
Technical Difficulties	Team members may encounter unfamiliarity with any tools and frameworks like React, Node.js or MySQL leading to Implement delays.	H	Members will use the first week of each iteration for learning and project setup and use resources like YouTube, Stack Overflow and official documentation for support and seeking help from the lecturer.
GitHub Conflicts / Code Loss	Continuation of edits by multiple teams might cause problem merge conflicts or accidental file overrides that might result in lost work.	M	We will enforce a branch strategy where each member will work on a separate branch and submits pull request for review before merging to main and Commit changes at least once daily.
Deadline Overrun	There might be delays in task and may take longer than estimated that might cause the team to miss interim or final submission deadlines.	M	Each member will have a 2-day build buffer before major deadlines that track progress weekly and using the Gantt Chart. And inform the team leader if any task is delayed by more than one day.
Communication Breakdown	Poor communication between team members may cause duplication of work, misaligned expectations or decisions.	L	There will be team meetings online or in-person using a application like WhatsApp and Discord groups for quick updates and Document all of the key decision in a shared notes file in GitHub.

Data Loss (Local Files)	Local Hardware failure like laptop crash could result in loss in project files, documentation or source code.	L	All Team members must push their code/document to GitHub at the end of their work sessions or any Critical documentation should be backed up in Online cloud storage services like Google drive, One Drive and Proton Drive etc.
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